MOBILE DEV · AI THAT RUNS ON THE PHONE

ARSENIUS PURBANDONO

I write Flutter apps for tax officers in West Java. Lately I've been teaching those apps to read license plates on the phone itself, no internet needed.

Bandung, Indonesia · West Java Revenue Agency (Bapenda Jabar) · since Feb 2023

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HOW I GOT HERE

`git log --reverse arsenius.` Oldest first.

0a1f3c2 SMKN 1 Simpang Empat

2016 – 2019 (edu)

Vocational high school, multimedia major.

3b7e9d0 Universitas Pendidikan Indonesia

2019 – 2023 (edu)

Information Systems & Technology Education (PSTI), Bandung.

5c21a8e HIMA PSTI

2021 (edu)

Chaired the legislative board of the department's student association.

CERTIFICATES

- **BNSP Senior Mobile Programmer** · 2025-11
- **BNSP Junior Mobile Programmer** · 2025-08
- **EF SET English B2** (54/100) · 2024-06
- MSIB Mobile & Cloud Dev, Nongsa Digital Park · 2022
- MSIB Game Development, Agate · 2021–22

7d40b15 MSIB · Agate

Aug 2021 – Jan 2022 (work)

Government-backed internship (MSIB, Kampus Merdeka), game development track.

8e93f6a MSIB · Nongsa Digital Park

Mar – Jul 2022 (work)

Government-backed internship (MSIB, Kampus Merdeka), mobile and cloud track.

5a1e922 Mobile Developer Intern · PT Safepedia

Aug – Dec 2022 (work)

Native Android in Kotlin for an app used by PT Antam's mining operators. Bug fixes and new features.

a76c0de Mobile Application Developer · Bapenda Jabar

Feb 2023 (work)

Joined West Java's revenue agency. Flutter apps for vehicle-tax tracing, Samsat counters and field checks, split into 4–19 packages, shipped through Shorebird and OTA. Still here.

b2c4e71 EF SET English B2

Jun 2024 (cert)

54/100.

c9f1d04 Community Manager · Web3 Developer Indonesia

Oct – Dec 2024 (work)

Ran the community side for a developer group, next to the day job.

d0a8b39 BNSP Mobile Programmer: Junior, then Senior

Aug → Nov 2025 (cert)

Indonesia's national certification, both levels in one year.

LANGUAGES

- Japanese: beginner. Working through Minna no Nihongo, aiming for JLPT N3
- English: B2 (EF SET 54/100)
- Indonesian: native

I want to work as a software or AI engineer in Japan. I can start in the second half of 2027 and will need visa sponsorship.

e57c2fa Started building my own plate models

Jan 2026 (ai)

ML Kit kept misreading Indonesian plates in the field. Began collecting data, wrote a labeling tool, trained a detector.

f13b8c6 Color classifier, then custom OCR in production

Oct – Nov 2026 (ai)

Panah Pasopati v4.5.0 dropped ML Kit. Exact plate reads went from 56% to 76%.

HEAD Learning Japanese, heading to Japan HEAD

now (now)

Working through Minna no Nihongo toward JLPT N3. Looking for a software or AI engineer role in Japan from the second half of 2027.

MOBILE IS THE DAY JOB

Every app below ships through this setup. Flutter first, Kotlin when a plugin can't do it.

ARCHITECTURE

Split into packages

4 to 19 Flutter packages per app: core, base, one per feature. One Dio client, go_router, Hive for local data, freezed models.

SHIPPING

Fixes without the Play Store wait

Shorebird patches for Dart changes, our own OTA for full APKs: HTTPS only, newer versions only, SHA-256 checked. I plan which fix can go out as a patch and which needs a new APK.

NATIVE

Down to Kotlin when needed

Patched the Android camera plugin so it stops restarting 3–4 times. Worked around Android 10+ stripping GPS from gallery photos.

HARDWARE

Not just phones

POS terminals at Samsat counters, self-service kiosks, thermal and Bluetooth printers, a second screen facing the customer.

SECURITY

Sessions and roles

Role checks at login and again per menu and route. Token refresh inside a Dio interceptor, a single 'session expired' dialog that logs the user out, device ID sent on login.

PERFORMANCE

Fast on field phones

On-device models run off the UI thread, so the crop step got 4× faster. Dropping ML Kit made the APK 30 MB smaller.

THE NUMBERS

3+
years writing Flutter at work

56→76%
plates read exactly right, ML Kit vs my model

−30MB
smaller APK after I removed ML Kit

4×
faster crop after moving it off the UI thread

1.1M
parameters in the OCR model (2.3 MB file)

494
real plates kept for testing, never trained on

HOW THE PHONE READS A PLATE

This runs inside Panah Pasopati. Everything happens on the phone. No server, no signal needed.

01
CAMERA
Frames from a cheap or mid-range Android phone

02
DETECT
YOLOv11n finds the plate and straightens it

03
CROP
Cropped once, in a background isolate

04
COLOR
MobileNetV2 tells white, black, yellow and red plates apart

05
READ
My CNN+CTC model reads the characters

06
LOOKUP
Number + color go to the vehicle tax lookup

APPS

Panah Pasopati

`Live · Flutter · TFLite · YOLO · Shorebird · Firebase · Kotlin`

Officers point the camera at a plate and get the vehicle's tax record. Used in the field across West Java.

- Built the plate-recognition module (Teman Pamor): detection, color, OCR, all on the phone
- Patched the Android camera plugin in Kotlin so the camera stops restarting 3–4 times
- Locked down OTA updates: HTTPS only, newer versions only, SHA-256 check
- Field rules: officers must be inside a location radius, a warning if the car was already traced, an owner-info form, and data scoped to each role's region
- Token refresh and forced logout when a session dies, plus a tracing report with a date filter

ATOS PAMOR

`Live · Flutter · MVC · Dio · Hive · flutter_map · Geolocator · OTA`

A fuller tracing app for the special field team: data checks, vehicle history and the paperwork for collecting overdue tax.

- Photo evidence gets a watermark with officer ID, GPS and the street address
- Every controller handles errors the same way now, and users see the real backend message
- Moved the app to a new major Flutter and Dart version
- Gallery photos lost GPS in the watermark (Android 10+ strips it). Shipped a fix that works as a Shorebird patch, and wrote down the full fix for the next APK
- Wrote the API integration and testing/debugging documents for ATOS PAMOR and Panah Pasopati

SAKTI JAWARA

`Live · Flutter · Device SDK · Thermal printer · Excel/PDF`

The counter app at every Samsat office in West Java. It runs on cashier-style POS terminals, where officers take tax payments and change vehicle data.

- Hooked up the device SDKs: camera, thermal printer, second screen
- Excel and PDF export
- Keyboard shortcuts, because counter staff hate reaching for the mouse

Sakti Mobile

`Live · Flutter · Camera · Excel/PDF`

SAKTI JAWARA, on a phone. Verifiers use it to check payments and data changes.

- Camera for evidence photos
- Local data storage
- Excel and PDF export

Samsat Digital

`Live · Flutter · Kiosk · Device SDK`

SAKTI JAWARA on a big self-service kiosk at Samsat branches, where people pay vehicle tax on their own.

- Connected the kiosk hardware: camera, printer and the rest
- Local data for each kiosk session

SEMAR

`Live · Flutter · Android + Web · Multi-package · Hive`

Water tax. Officers visit taxpayer companies and record their water-meter readings, then a verifier accepts or rejects each one.

- Split into packages: recording, verification and the two histories
- One codebase for Android and the web. Detail pages load by the ID in the URL, so refreshing the browser doesn't break them
- Menus by role (recorder, verifier, admin). Other roles are turned away at login
- Second evidence photo and notes on each reading, an update check, Shorebird patches, a server switcher and a debug logger

Pajak Alat Berat

`Live · Flutter · MVC`

Tax app for heavy equipment like excavators and cranes: recording and tracing.

- Features and maintenance on the shared Flutter setup

Aing Linmas

Prototype · Flutter · Android + iOS · QR · PDF

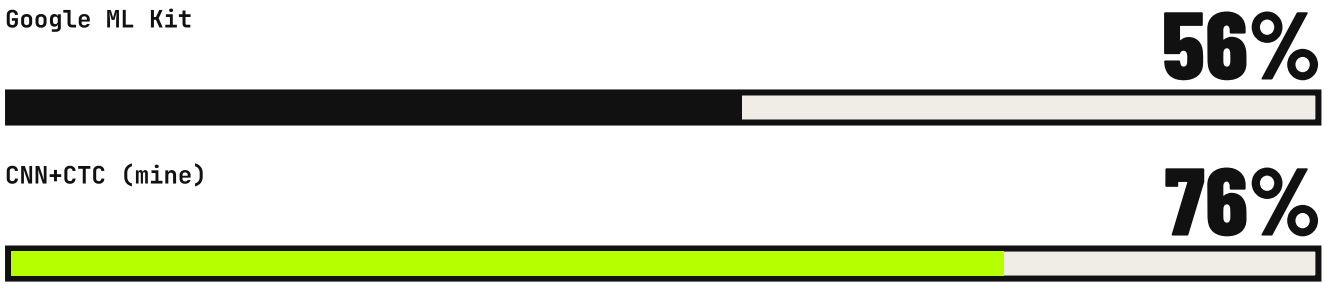
Phone version of a sign-up system for Linmas community-protection members, from province down to village.

- 3-step sign-up with an ID-number check and drafts that save themselves
- Digital member card with a QR code, saves as image or PDF
- Admin side: recap dashboard by region, verify or reject with a reason, scan a card's QR to check it
- Runs on fake data for now, no backend yet

MODEL LAB

The plate-recognition repo: data, labels, training. 494 test plates were set aside before training and never used to tune anything, so the accuracy numbers are honest.

WHY I DROPPED GOOGLE ML KIT



ML Kit is a general OCR. On Indonesian plates it kept mixing up O and 0, so officers had to fix the result by hand.

First I set aside 494 real plates as a test set and split by plate, so the same car never shows up in training. Then I generated 30k synthetic plates in the plate font, trained a small CNN+CTC model and exported it to TFLite, checking that the TFLite model gave the same results as the original.

It only went into the app after it beat ML Kit on that test set: 76% vs 56% exact match. Dropping ML Kit also took about 30 MB off the APK.

Plate detector

LIVE

YOLOv11n → TFLite

mAP50 81.6%

Finds the plate and fixes the rotation when the photo is taken at an angle.

Plate OCR

LIVE

CNN + CTC · 1.1M params · 2.3 MB

76% exact

Reads the characters. ML Kit got 56% on the same 494 plates.

Plate color

LIVE

MobileNetV2 α0.35 · 96×96

~97% val acc

White, black, yellow, red. Red is still weak: not many real red plates to train on.

SANDBOX

Side projects I did to learn NLP and transfer learning.

Amazon Review Sentiment

87.25% acc · F1 0.87

TF-IDF · Logistic Regression · NLTK · Flask

1.6M reviews to train, 400k to test.

Casting Defect Inspection

91.8% val acc

MobileNetV2 transfer learning · Keras · Tkinter

Good vs defective pump impellers, 1,016 factory photos.

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